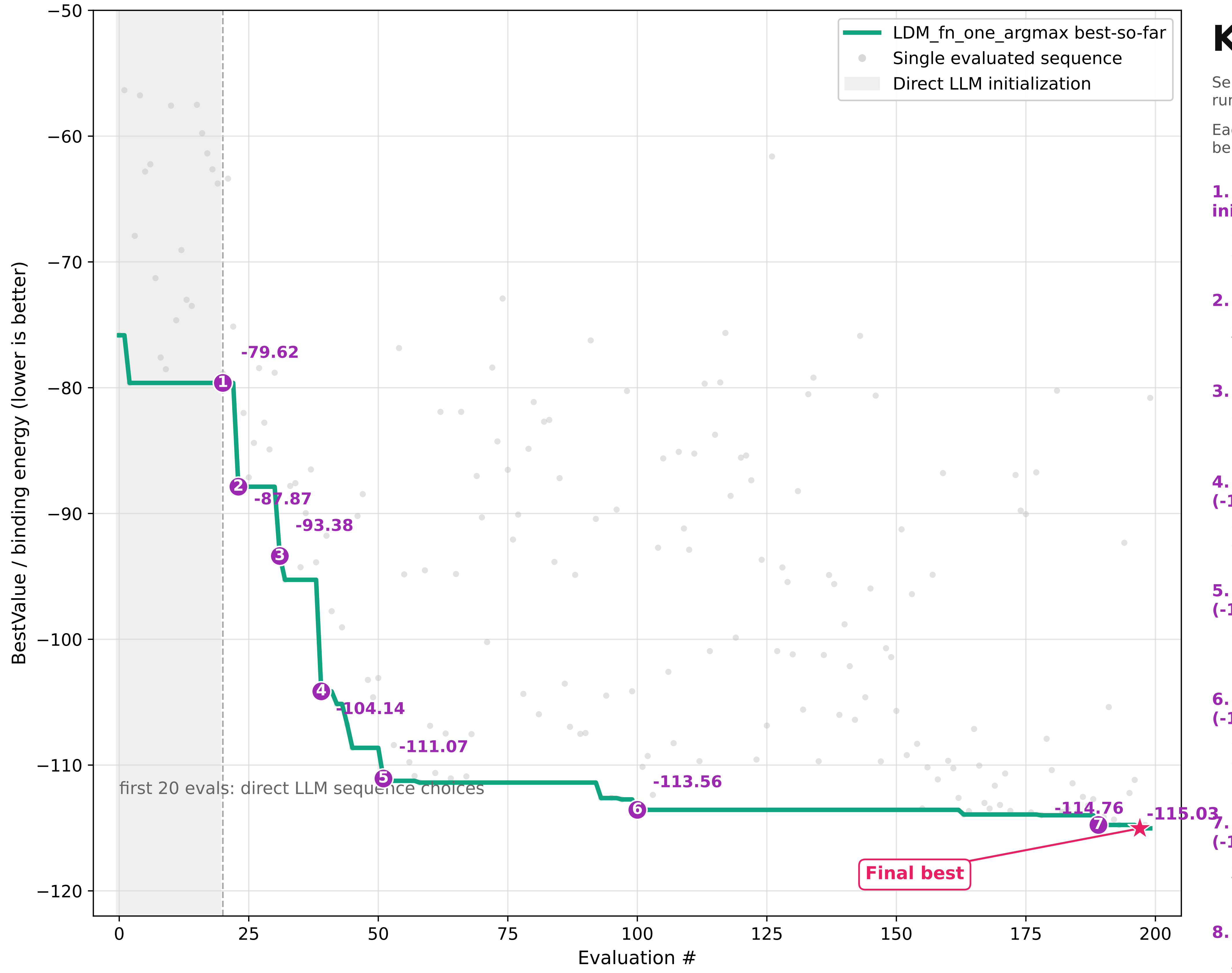

Key events

Selected run: best final value among 25 formal LDM_fn_one_argmax-style runs; also best among the five 1FBI_X seeds.

Each event lists the best 11-aa antibody at that point and the search behavior associated with the improvement.

1. LDM sets the first search region after initialization stall (-79.62)

Antibody: `IFYEMAPDKHG`

After direct LLM initialization stops improving, LDM updates the search setup and the BO loop evaluates candidates in that region.

2. LDM finds the first large improvement (-87.87)

Antibody: `IFYEMKPNWHI`

The first post-initialization search region produces a clear improvement, moving the best value from -79.62 to -87.87.

3. LDM moves into a better motif family (-93.38)

Antibody: `ILYAMKFNWLP`

LDM moves the search to a different local family, and BO discovers a stronger sequence.

4. LDM escapes the plateau with a wider search move (-104.14)

Antibody: `ILYIFKFNYFI`

After a plateau near -95.27, LDM changes the search setup instead of continuing the same local pattern.

5. Focused LDM exploitation creates the main drop (-111.07)

Antibody: `ILLLFGFNLFI`

Repeated local searches around related high-scoring variants turn the early gains into a much stronger best value.

6. LDM returns to a strong region and improves again (-113.56)

Antibody: `FLLLLCFNLFI`

After many weak evaluations, LDM brings the search back to a high-performing region and extracts a smaller but meaningful improvement.

7. Near-best variants are refined after a long stall (-114.76)

Antibody: `LLLFFCFNLFF`

The run spends many evaluations near a plateau, then a focused refinement step improves the best value again.

8. Final best found (-115.03)

Antibody: `LLLFFCLHLFF`

The last local refinement produces the final best value.